

From design to chip

Some of the things you can do with this tool are digital filter architectural layouts, printed circuits displaying up to eight layers and artwork for your graphic design. The schematic can be a digital or an analogue circuit. As you work, information on connectivity and geometry in regions is carefully stored to pull back the same while making your design checks. Electric provides options for checking electrical rules in your design like well areas, proper contact and spacing, and to spot antenna violations. Confirming compatibility with design rules is easy via the incremental checker that displays errors when a spacing, notch or sizing rule is violated.

A highly-flexible simulator

A built-in simulator called ALS can perform 12-state switch-level simulations. On simulation, a waveform display lets you access different parts of the circuit and observe their functioning well. You can also produce input decks for a number of popular simulators like device-level simulators such as Spice, switch-level ones like Silos, Tegas, IRSIM, ESIM, RSIM, RNL, Cosmos and Mossim, and even miscellaneous simulators such as PALs.

For the layout and the like

You can route your layout components using four different options: maze router for connecting a single wire between points, cell stitching for making explicit connections in overlapping areas and river router for multiple parallel wire channels between cells. If you are lazy, there is an easy way out—the mimic router. It simply watches you manually create or delete a wire the first time after it is called and repeats the same activity in similar areas all around

the circuit. Sit back and watch the circuit kick into action!

There is a logical effort system for marking digital schematic gates with fan-out information that you can use to produce optimally-fast circuits. You can work with customised arrays, thanks to CMOS PLA generator. If you are having trouble with standard cells, call Silicon Compiler to take care of the placing and routing between these cells with the help of a structural netlist.

Getting a layout for your memory is easy using ROM generator, which automatically creates one

sense, this means that the physical layout you design decides the fate of your chip and, thus, it is very important that every design aspect is correctly represented.

The database on Electric attaches geometric data with every associated component and wire to ensure the above fact. This database is built on network structure with nodes corresponding to components and arcs to connections, resulting in a well-connected structure.

Also, the database is made extendable to allow for storing additional structures describing param-

Taking a look-back

- Written in 1982 by Steven M. Rubin at Fairchild A.I. Laboratory in Palo Alto, California, USA
- Under Free Software Foundation since 1998
- Initially developed in C language; transition to Java began in 2003
- Managed by Static Free Software, now a part of RuLabinsky Enterprises Inc.
- Works on Mac, Unix and Windows systems

from the personality table. Check for network consistency using the available checker. It uses Gemini algorithm to compare a layout with its original schematic, so you can be sure that the way you created your layout did not alter the circuit basics.

Once the layout is complete, finish it off with pad cells by placing it using Pad Frame generator, which even completes the wiring around the core.

The idea behind, the strong backbone

Simply stated, Electric combines graphics, connectivity and accurate geometry to create a single, highly-flexible tool to create your design. Complete control over all the above parameters is a huge task. With this in mind, the tool is structured based on four main aspects, which we will see next.

Representation. At the end of it all, it is the layout you create for your circuit that gets converted into the hardware chip. In a

ters like behaviour, power consumption, design rules and such.

Programmability. Electric uses the concept of constraints effectively to give you an easy and interesting programming experience. You specify layout relationships via the constraint system, which hierarchically propagates through the design and is continuously enforced at all stages. With these constraints being spatial in nature, even more programming flexibility is introduced by supporting interpretive systems like LISP, TCL and Mathematica for some parts of the system.

Design environment. With the environment varying from ever so minutely to involving massive change, design environment is unlimited. Electric accepts this fact and works to include these in a uniform manner. You can express the surrounding in network form with all graphic attributes and simulation behaviour, create a module and integrate this with the system.